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Sreeetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

C++ CAPSTONE PROJECT PRESENTATION

DIGITAL LIBRARY & PAPERLESS LEARNING RESOURCE MANAGEMENT SYSTEM

GOWTHAM S
1971260007

SIDHARDHA V
1965260034

GOKUL K
1911260026

Course name : **Object Oriented Programming concepts in C++**

Course code : **DSA0101**

Guided by : **Dr. K. PARIMALA KANAGA DEVAN**



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Introduction

Modern Academic Need

Higher education institutions are transitioning rapidly from physical print media to digitized learning assets. Managing digital books, e-journals, lecture media, and research repositories requires systematic access control and streamlined digital indexing.

Paperless Ecosystem

A paperless learning resources system eliminates manual paperwork, physical ledger logging, and lost book entries. It facilitates instant digital checkouts, continuous availability monitoring, and eco-friendly institutional resource administration.

Literature Sumey (2024-2025)

Recent innovations, gaps identlied, and our proposed augmentations.

Author	Year	Method	Pros / Cons	Our SOLItiofl
Lee	2025	Oline-first reader app	Pros: accessibil•cy. Cons: sync conflicts	Conflict–resilient sync whh versioning
Nguyen & Silva	2025	Open-access analyics	Pros: transparency. Cons: privacy concerns	Privacy–preserving aggregated analyics
Kumar et al.	2024	AI–based content tagging	Pros: faster indexing. Cons: bias in models	HUlzlan–in–loop V ïdation + fairness checks

System Objectives



Automate Operations

Develop a software core to handle book issuance, return tracking, member authentication, and catalog searches seamlessly.



Digital Cataloging

Implement structured data models for digital resources including eBooks, journal PDFs, lecture notes, and media files.



Paperless Governance

Enforce file-based dynamic storage, rule-based user access, and operational transparency across all administrative tiers.

Existing vs. Automated System

Legacy Manual System

- Physical ledger registers for book borrowing and returns.
- Manual card catalog searching leading to slow discovery.
- High risk of physical book damage, theft, and loss.
- Zero real-time metric updates for library managers.

Proposed Digital System

- Instant console-based OPAC search by title, author, or tag.
- Automated license management & concurrency tracking.
- Persistent binary/text file data logging for zero data loss.
- Automated analytics on borrowing trends and user access.



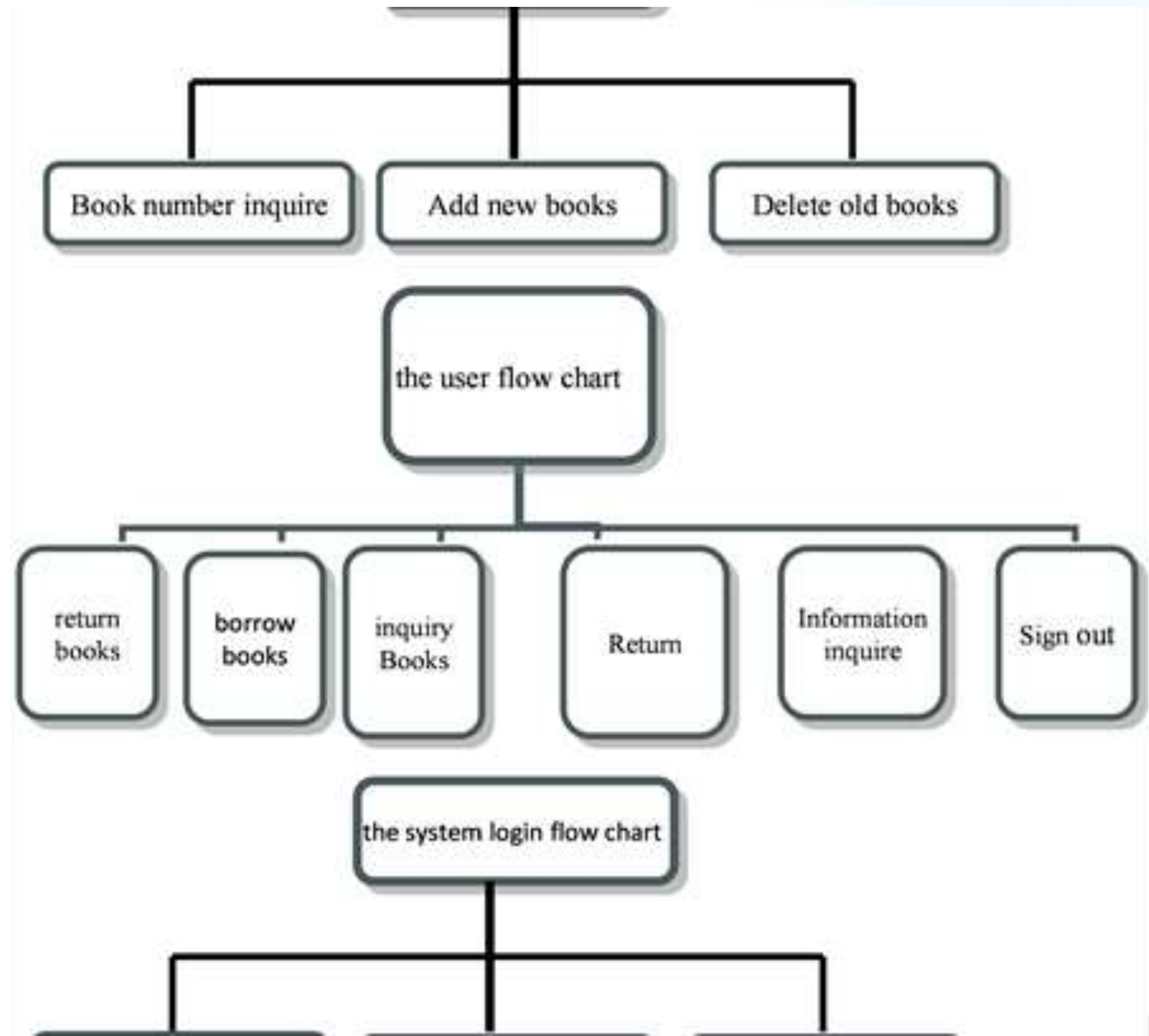
Proposed C++ Architecture

The proposed Digital Library & Paperless System offers a lightweight, high-performance console application engineered with clean Object-Oriented Principles.

Role-Based Access: Distinct views for Students, Faculty, and System Administrators.

File Processing Engine: Saves catalog state and transactional logs continuously to disk.

Resource Indexer: Fast lookup using dynamic vectors and array search algorithms.





System Architecture & Modules

Module 1: User Mgmt

Handles student/faculty accounts, user privileges, and active profile records.

Module 2: Classification

Auto-categorizes resources by domain, media type, tags, and license type.

Module 3: Resource Engine

Maintains eBook inventories ,file URLs, copies available, and access parameters.

Module 4: Access Control

Processes checkouts, active download holds, issue limits, and return schedules.

Module 5: Archiving

Organizes course readers ,digital archives, and paperless document repositories.

Module 6: Admin Reports Generates

usage stats,system errorlogs, overdue reports, and summary metrics.

Module 1 - User Management

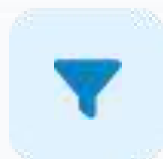
 **Account Creation:** Register new students, faculty members, and library staff with unique credentials.

 **Role Authentication:** Enforces login validation for Admin privileges vs standard patron rights.

 **Member Profile Tracking:** Stores current borrowed digital items, download limits, and fine history.



Module 2 - Smart Classification



Category Tagging

Classifies resources into Computer Science, Engineering, Mathematics, Humanities, and General Science domains.



Format Indexing

Tracks media types including PDF eBooks, EPUB readers, audio lectures, and interactive code notebooks.



Search Filtering

Implements fast string comparison methods to filter by ISBN, author, keywords, or title substrings.

Module 3 - Digital Resource Engine

Resource Inventory Logic

Maintains dynamic records of all available learning materials.

Tracks current stock, allocated active licenses, digital file paths, and publisher metadata in real time.

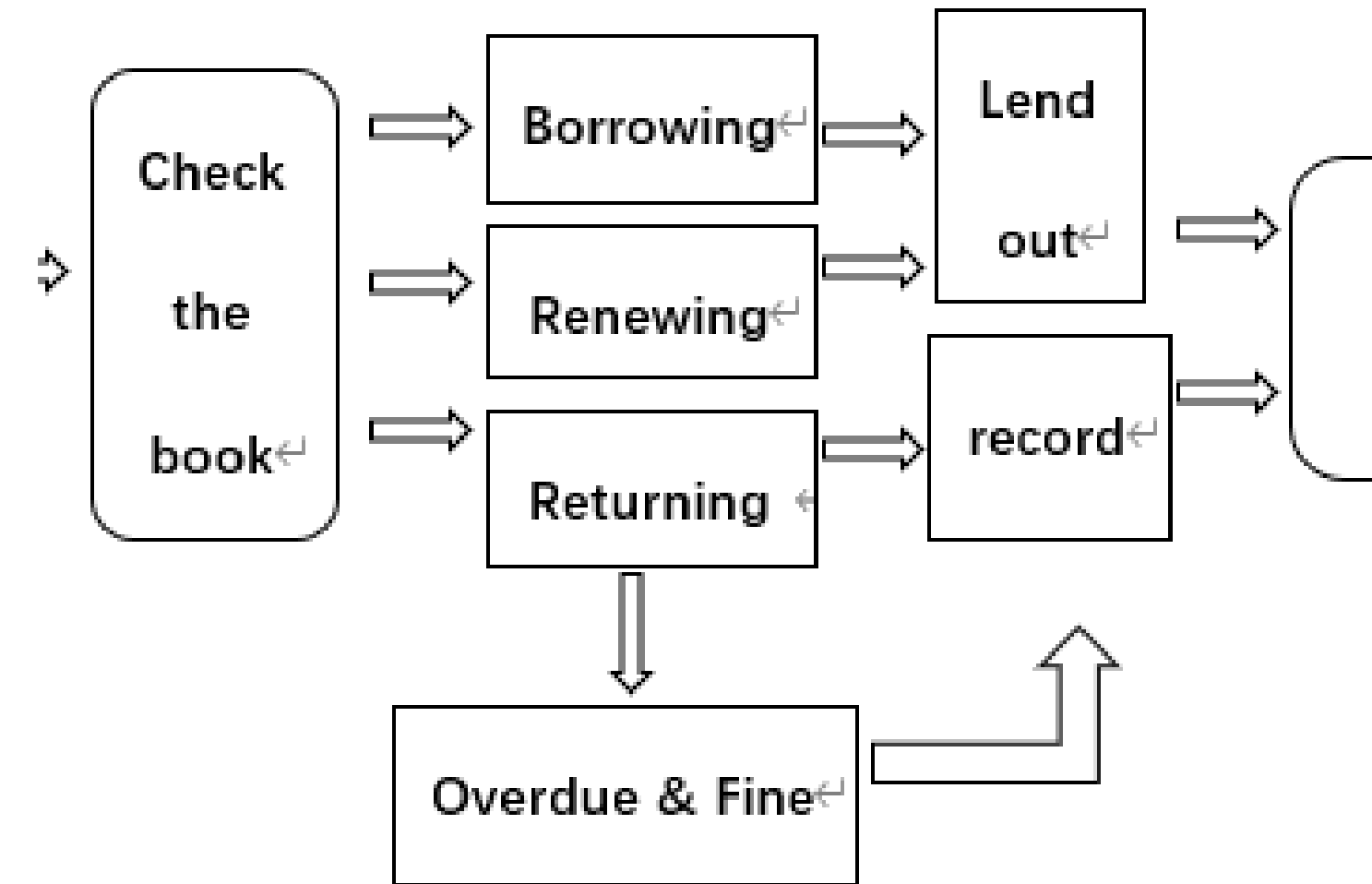
CRUD Operations

Provides authorized administrators with options to add newly acquired digital titles, update existing metadata, remove deprecated editions, and modify active license limits.



Module 4 - Borrowing & Access Control

- ➔ **Automated Digital Checkout:** Users can request instant access keys or downloadable copies if licenses are within safe thresholds.
- ⌚ **Loan Duration Enforcement:** Sets auto-expiring checkout periods (e.g., 14 days for students, 30 days for faculty).
- 📄 **Fine Calculation Logic:** Programmatically calculates overdue fines for delayed digital returns based on daily penalty rates.
- 📋 **Waitlist Prioritization:** Queues checkout requests when maximum digital license limits are temporarily reached.



Module 5 - Archiving & Document Control

Paperless Repository Features

Course Reader Archiving: Bundles subject material, syllabus PDFs, and historical exam papers per department.

Disk Storage Compression: Efficient file record formatting in binary stream format (`.fstream`).

Version Control: Maintains old textbook editions in secondary archive storage for historical research.



Module 6 - Admin Analytics & Metrics

RESOURCE ACCESS DISTRIBUTION BY MEDIA TYPE (MONTHLY DOWNLOADS)

eBooks / Textbooks

85%

Research Journals

65%

Lecture Slide Decks

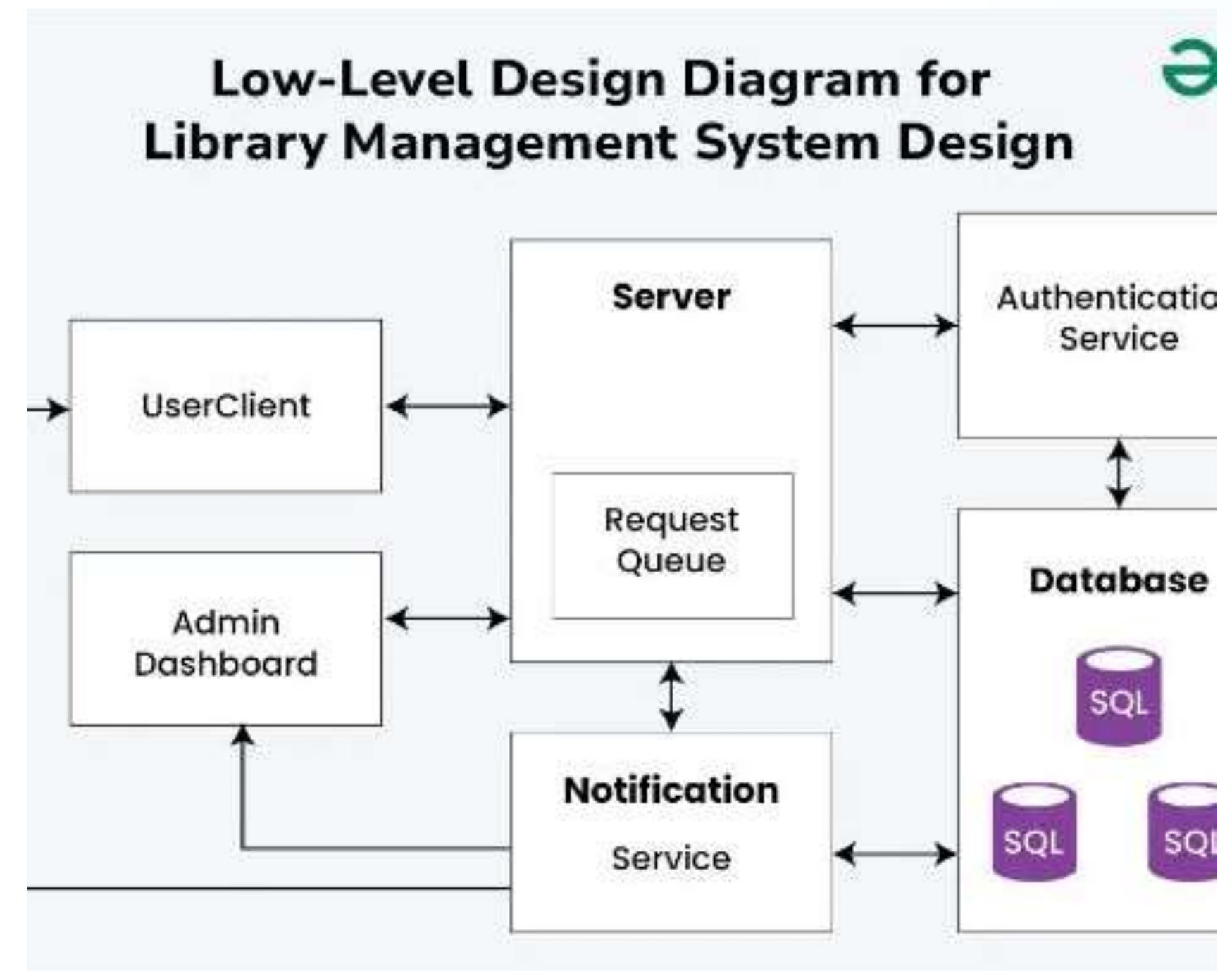
50%

Past Exam Papers

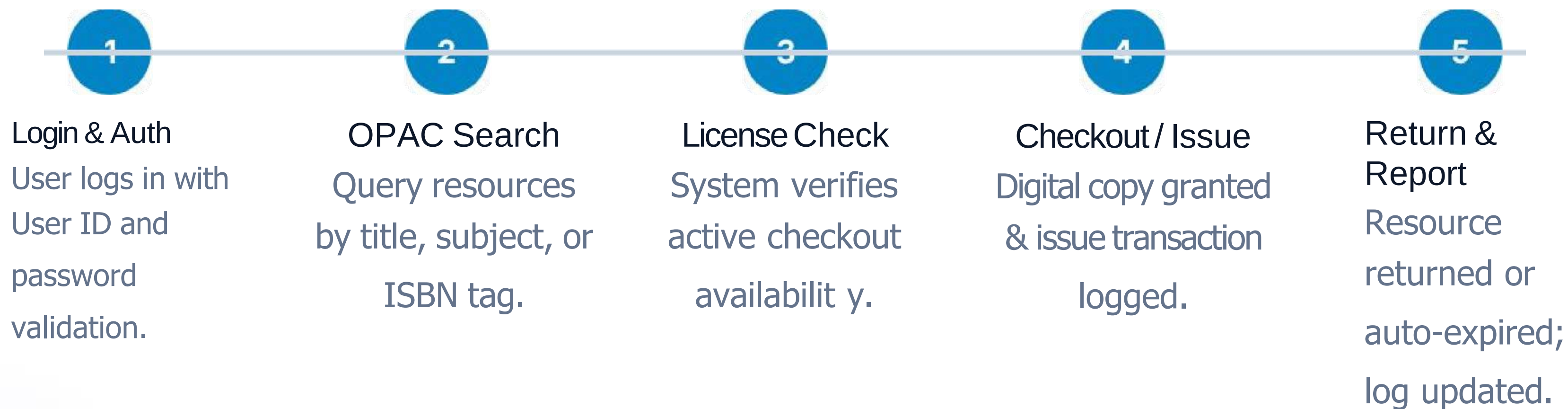
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Use Case & System Interaction

-  **Students:** Search catalog, checkout eBooks, read course materials, view active due dates.
-  **Faculty:** Recommend reading lists, reserve digital textbooks, request custom archives.
-  **Librarian / Admin:** Add/modify catalog inventory, manage member privileges, extract system logs.



System Workflow

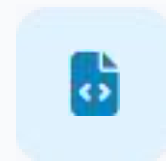


C++ Programming Concepts Applied



OOP Principles

Utilizes Classes, Inheritance, Encapsulation, and Polymorphism for clean separation of concerns.



File Handling

Uses `ifstream` and `ofstream` stream objects for persistent storage of user profiles and catalog lists.



STL Containers

Applies `std::vector` and standard algorithms for dynamic-memory allocation and efficient data sorting.



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System Advantages

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secure	Instant OPAC Search	Resource Accessibility	Third-party Software Cost

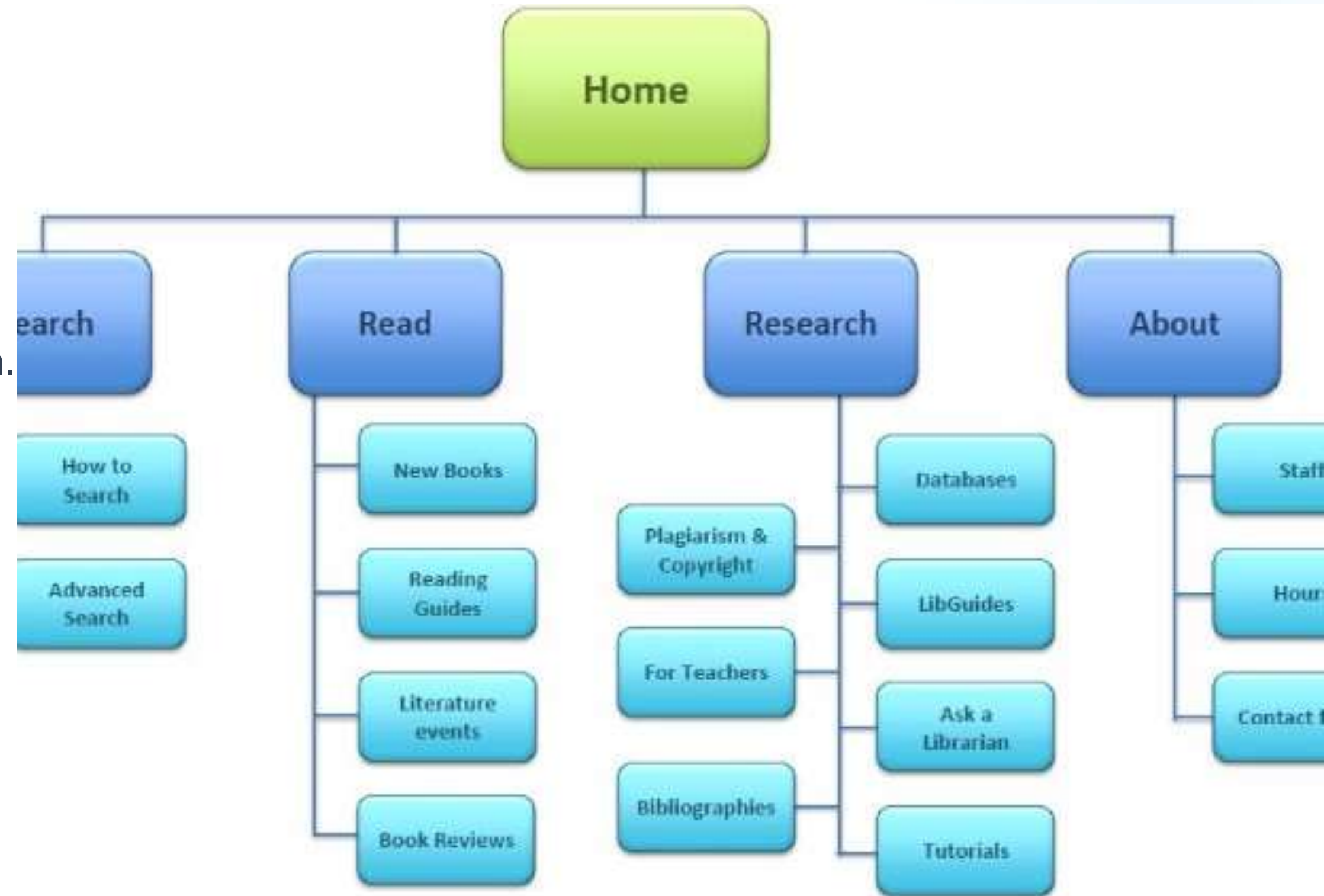
- ⚡ **High Operational Speed:** compilation ensures fast execution and low memory footprint.
- ♻️ **Eco-Friendly & Sustainable:** Reduces paper consumption across academic departments.
- 🔧 **Modular Maintenance:** Clean modular design allows seamless expansion of new features.

Future Enhancements

☁ **Cloud Storage Syncing:** Integrate remote S3/GCS buckets for unlimited asset distribution.

🧠 **AI Semantic Search:** Incorporate natural language processing for smart vector-based resource recommendations.

📱 **Cross-Platform Mobile App:** Build flutter/React Native frontends interacting with C++ backend services.





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Conclusion

The Digital Library & Paperless Learning Resource Management System demonstrates how foundational Object-Oriented paradigms and file processing techniques can effectively automate academic resource administration, reduce paper waste, and deliver an intuitive learning platform.

Thank You!

<https://github.com/sidhardhareddy339-ops/DSA--oop>

 **Capstone Project** |

 **Digital Library Management System**